

MATH I - Week 1: NOTES 2

on Positional representations of numbers, Scientific and Engineering Notations

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Positional representation of numbers

A **number** is an abstract mathematical object, whereas a **numeral** is a written representation of that number. For example, the same number can be represented as

$$13_{10} = 1101_2 = D_{16} = XIII.$$

The first three are *positional* representations in different bases; XIII is a *non-positional* Roman-numeral representation.

1. Positional numeral systems

In a **positional numeral system**, the value of a digit depends on:

1. the digit itself;
2. its position;
3. the base of the system.

For example, in decimal notation,

$$(3527)_{10} = 3(10^3) + 5(10^2) + 2(10^1) + 7(10^0).$$

Therefore,

$$3527 = 3000 + 500 + 20 + 7.$$

The digit 5 represents 500, not simply 5, because of its position.

2. Common positional systems

- System: Binary; base: 2; digits: 0, 1; example: 1101_2
- System: Octal; base: 8; digits: 0, ..., 7; example: 15_8
- System: Decimal; base: 10; digits: 0, ..., 9; example: 13_{10}
- System: Hexadecimal; base: 16; digits: 0, ..., 9, A, ..., F; example: D_{16}

In hexadecimal notation,

$$A = 10, \quad B = 11, \quad \dots, \quad F = 15.$$

All the following numerals represent the same number:

$$1101_2 = 15_8 = 13_{10} = D_{16}.$$

For example,

$$1101_2 = 1(2^3) + 1(2^2) + 0(2^1) + 1(2^0) = 8 + 4 + 0 + 1 = 13_{10}.$$

3. The importance of zero

Zero serves as a **placeholder** in positional systems. For example,

$$507 = 5(10^2) + 0(10^1) + 7(10^0).$$

The zero shows that there are no tens. Without it, 507 could not be distinguished from 57.

Similarly,

$$10101_2 = 1(2^4) + 0(2^3) + 1(2^2) + 0(2) + 1 = 21_{10}.$$

4. Non-positional representations

In a **non-positional system**, a symbol generally has a fixed value that does not depend primarily on its position.

Roman numerals

Roman numerals use symbols such as

$$I = 1, \quad V = 5, \quad X = 10, \quad L = 50, \quad C = 100.$$

For example,

$$13 = XIII.$$

Roman numerals are useful for labels, dates and numbering sections, but arithmetic is considerably less convenient than in positional notation.

See an example on how difficult are non-positional systems such as the Roman numbers to represent arithmetic:

Example of the difficulty of using Roman numbers:

With the help of the Roman numeral converter:

<https://www.calculatorsoup.com/calculators/conversions/roman-numeral-converter.php>

(<https://www.calculatorsoup.com/calculators/conversions/roman-numeral-converter.php>) we get:

$$2021 = MMXXI$$

and

$$9999 = MMMMMMMMMCMXCIX$$

which is too lengthy.

However, it can become much more lengthy if we add one more decimal digit to the 9999 number:

$$99998 =$$

MMMMMMMMMMMMMMMMMMMM
MMMMMMMMMMMMMMMMMMMM
MMMMMMMMMMMMMMMMMMMM
MMMMMMMMMMMMMMMMMMMM
MMMMMMMMMMMMMMMMMMMM
MMMMMMMMMMMMCMLXCVIII

5. Why computers use binary numerals

Electronic devices can reliably distinguish between two physical states, such as:

low/high voltage, off/on, 0/1.

Computers therefore represent information using **bits**, and binary is their fundamental positional numeral system.

The essential point to remember is:

A positional system determines how a number is written—not what the number is.

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Tutorial: on Scientific and Engineering Notation

Scientific and engineering notation provide compact ways to write extremely large or small numbers. They are especially useful in physics and engineering, where quantities range from subatomic dimensions to astronomical distances.

For example, the speed of light can be written as

$$299\,792\,458\text{ m/s} = 2.99792458 \times 10^8\text{ m/s}.$$

1. Powers of ten

A power of ten indicates how many times 10 is multiplied or divided:

$$\begin{aligned} 10^3 &= 1000, & 10^2 &= 100, & 10^1 &= 10, & 10^0 &= 1, \\ 10^{-1} &= 0.1, & 10^{-2} &= 0.01, & 10^{-3} &= 0.001. \end{aligned}$$

A positive exponent produces a large value, while a negative exponent produces a small value:

$$10^6 = 1\,000\,000, \quad 10^{-6} = 0.000001.$$

2. Scientific notation

A number is in **scientific notation** when it is written as

$$\boxed{a \times 10^n},$$

where

$$1 \leq |a| < 10$$

and n is an integer.

The number a is sometimes called the **coefficient** or **significand**, and n is the exponent.

Large-number example

$$4\,500\,000 = 4.5 \times 10^6.$$

The decimal point moves six positions to the left, so the exponent is +6.

Small-number example

$$0.000072 = 7.2 \times 10^{-5}.$$

The decimal point moves five positions to the right, so the exponent is -5 .

Negative-number example

$$-83\,000 = -8.3 \times 10^4.$$

The negative sign does not affect the exponent.

3. Converting scientific notation to decimal notation

A positive exponent moves the decimal point to the right:

$$3.27 \times 10^4 = 32\,700.$$

A negative exponent moves it to the left:

$$3.27 \times 10^{-4} = 0.000327.$$

4. Engineering notation

A number is in **engineering notation** when it has the form

$$\boxed{a \times 10^{3k}},$$

where $k \in \mathbb{Z}$ and normally

$$1 \leq |a| < 1000.$$

The exponent must therefore be a multiple of three:

$\dots, -12, -9, -6, -3, 0, 3, 6, 9, 12, \dots$

For example,

$$4.7 \times 10^7$$

is valid scientific notation but not engineering notation because 7 is not a multiple of 3. In engineering notation, it becomes

$$\boxed{47 \times 10^6}.$$

Similarly,

$$2.5 \times 10^{-5} = \boxed{25 \times 10^{-6}}.$$

5. Scientific versus engineering notation

Consider the number

0.0000475.

In scientific notation,

$$\boxed{4.75 \times 10^{-5}}.$$

In engineering notation,

$$\boxed{47.5 \times 10^{-6}}$$

The two expressions represent exactly the same number.

Feature	Scientific notation	Engineering notation
General form	$a \times 10^n$	$a \times 10^{3k}$
Coefficient	$1 \leq a < 10$	$1 \leq a < 1000$
Exponent	Any integer	Multiple of 3
Main use	Science and general mathematics	Engineering and SI prefixes
Example	4.7×10^7	47×10^6

Engineering notation is particularly convenient because powers of 10^3 correspond to SI prefixes.

6. Engineering notation and SI prefixes

Power	Prefix	Symbol	Example
10^{12}	tera	T	terawatt, TW
10^9	giga	G	gigahertz, GHz
10^6	mega	M	megawatt, MW

Power	Prefix	Symbol	Example
10^3	kilo	k	kilovolt, kV
10^0	—	—	volt, metre, second
10^{-3}	milli	m	millimetre, mm
10^{-6}	micro	μ	microampere, μA
10^{-9}	nano	n	nanometre, nm
10^{-12}	pico	p	picofarad, pF
10^{-15}	femto	f	femtosecond, fs

For example,

$$25 \times 10^3 \text{ V} = 25 \text{ kV},$$

$$4.7 \times 10^{-6} \text{ A} = 4.7 \mu\text{A},$$

and

$$220 \times 10^{-12} \text{ F} = 220 \text{ pF}.$$

Notice that uppercase and lowercase letters matter:

$$\text{m} = 10^{-3}, \quad \text{M} = 10^6.$$

Thus, 5mW and 5MW are very different:

$$5 \text{ mW} = 5 \times 10^{-3} \text{ W},$$

$$5 \text{ MW} = 5 \times 10^6 \text{ W}.$$

They differ by a factor of 10^9 .

7. Important scientific and engineering quantities

Speed of light

The speed of light in vacuum is exactly

$$c = 299\,792\,458 \text{ m/s}.$$

Scientific notation:

$$c = 2.99792458 \times 10^8 \text{ m/s}.$$

Engineering notation:

$$c = 299.792458 \times 10^6 \text{ m/s}$$

or approximately

$$c \approx 300 \text{ Mm/s}.$$

Elementary electric charge

The magnitude of the charge of an electron is

$$e = 0.0000000000000000001602176634 \text{ C}.$$

Scientific notation:

$$e = 1.602176634 \times 10^{-19} \text{ C}.$$

Engineering notation:

$$e = 160.2176634 \times 10^{-21} \text{ C}.$$

Avogadro constant

The number of specified particles in one mole is

$$N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}.$$

In engineering notation,

$$N_A = 602.214076 \times 10^{21} \text{ mol}^{-1}.$$

Planck constant

Planck's constant is

$$h = 6.62607015 \times 10^{-34} \text{ J s}.$$

In engineering notation,

$$h = 662.607015 \times 10^{-36} \text{ J s}.$$

For very small quantities like this, scientific notation is often clearer than using an uncommon SI prefix.

Standard atmospheric pressure

Standard atmospheric pressure is

$$101\,325 \text{ Pa}.$$

Scientific notation:

$$1.01325 \times 10^5 \text{ Pa}.$$

Engineering notation:

$$101.325 \times 10^3 \text{ Pa} = 101.325 \text{ kPa}.$$

Approximate mean radius of Earth

The mean radius of Earth is approximately

6 371 000 m.

Scientific notation:

$$6.371 \times 10^6 \text{ m}.$$

Engineering notation:

$$6.371 \text{ Mm} = 6371 \text{ km}.$$

In ordinary engineering practice, 6371 km is more familiar than 6.371 Mm.

Diameter of a human hair

A typical human hair might have a diameter of approximately

0.000070 m.

Scientific notation:

$$7.0 \times 10^{-5} \text{ m}.$$

Engineering notation:

$$70 \times 10^{-6} \text{ m} = 70 \mu\text{m}.$$

Visible-light wavelength

Green light has a wavelength of approximately

0.000000550 m.

Scientific notation:

$$5.50 \times 10^{-7} \text{ m}.$$

Engineering notation:

$$550 \times 10^{-9} \text{ m} = 550 \text{ nm}.$$

Engineering notation is especially useful here because wavelengths are commonly expressed in nanometres.

Nanotechnology scale

A feature measuring 25 nanometres has size

$$25 \text{ nm} = 25 \times 10^{-9} \text{ m}.$$

Scientific notation:

$$2.5 \times 10^{-8} \text{ m}.$$

Engineering notation:

$$25 \times 10^{-9} \text{ m} = 25 \text{ nm}.$$

Power-engineering examples

A power station producing 500 000 000 W has an output of

$$5.00 \times 10^8 \text{ W}$$

in scientific notation and

$$500 \times 10^6 \text{ W} = 500 \text{ MW}$$

in engineering notation.

A transmission voltage of 220 000V can be written as

$$2.20 \times 10^5 \text{ V}$$

or, more naturally,

$$220 \times 10^3 \text{ V} = 220 \text{ kV}.$$

Electrical and electronic components

A capacitor may have capacitance

$$0.000000000220 \text{ F}.$$

Scientific notation:

$$2.20 \times 10^{-10} \text{ F}.$$

Engineering notation:

$$220 \times 10^{-12} \text{ F} = 220 \text{ pF}.$$

A resistor with resistance

$$4\,700 \, \Omega$$

is written as

$$4.7 \times 10^3 \, \Omega = 4.7 \text{ k}\Omega.$$

8. Arithmetic in scientific notation

Multiplication

Multiply the coefficients and add the exponents:

$$(a \times 10^m)(b \times 10^n) = ab \times 10^{m+n}.$$

For example,

$$(3 \times 10^5)(2 \times 10^{-3}) = 6 \times 10^2 = 600.$$

Division

Divide the coefficients and subtract the exponents:

$$\frac{a \times 10^m}{b \times 10^n} = \frac{a}{b} \times 10^{m-n}.$$

For example,

$$\frac{8 \times 10^7}{2 \times 10^3} = 4 \times 10^4.$$

Addition and subtraction

Before adding or subtracting, express the numbers using the same power of ten:

$$3.2 \times 10^5 + 4.5 \times 10^4.$$

Rewrite the second number:

$$4.5 \times 10^4 = 0.45 \times 10^5.$$

Therefore,

$$3.2 \times 10^5 + 0.45 \times 10^5 = \boxed{3.65 \times 10^5}.$$

Powers

$$(a \times 10^n)^m = a^m \times 10^{nm}.$$

For example,

$$(2 \times 10^3)^2 = 4 \times 10^6.$$

9. Significant figures

Scientific notation clearly communicates the intended precision of a measurement.

For example,

$$3 \times 10^5$$

has one significant figure, while

$$3.000 \times 10^5$$

has four significant figures. Although both have the same nominal value, the second suggests greater measurement precision.

Suppose a length is measured as

$$L = 2.35 \text{ m.}$$

This value has three significant figures. Writing

$$L = 2.350000 \text{ m}$$

would incorrectly suggest a much more precise measurement unless that precision was actually available.

10. Calculator and computer notation

Calculators and programming languages often use E or e notation:

$$2.99792458\text{E}8$$

means

$$2.99792458 \times 10^8.$$

Similarly,

$$1.602176634\text{e}-19$$

means

$$1.602176634 \times 10^{-19}.$$

Here, e means “multiply by 10 raised to the following exponent.” It does **not** mean Euler’s number $e \approx 2.71828$.

In-Class Activity:

Try to solve the following problems. Keep your solutions and check them with the answers provided later (and shown below). The in-class activity is to facilitate your learning; it will not be graded.

Exercise 1

Write 0.00000824 in scientific and engineering notation.

Exercise 2

Write $7.35 \times 10^8 \text{ Hz}$ in engineering notation.

Exercise 3

Write 0.0047 F using an SI prefix:

11. Worked conversion examples

Exercise 1

Write 0.00000824 in scientific and engineering notation.

Scientific notation:

$$0.00000824 = \boxed{8.24 \times 10^{-6}}.$$

Since -6 is a multiple of 3, this is also valid engineering notation:

$$\boxed{8.24 \times 10^{-6} = 8.24 \mu}.$$

The prefix must be attached to an appropriate unit—for example,

$$8.24 \times 10^{-6} \text{ A} = 8.24 \mu\text{A}.$$

Exercise 2

Write $7.35 \times 10^8 \text{ Hz}$ in engineering notation.

Rewrite the exponent as a multiple of 3:

$$7.35 \times 10^8 = 735 \times 10^6.$$

Therefore,

$$\boxed{7.35 \times 10^8 \text{ Hz} = 735 \text{ MHz}}.$$

Exercise 3

Write 0.0047 F using an SI prefix:

$$0.0047 \text{ F} = 4.7 \times 10^{-3} \text{ F} = \boxed{4.7 \text{ mF}}.$$

Summary

Scientific notation requires one nonzero digit before the decimal point:

$$\boxed{1 \leq |a| < 10}.$$

Engineering notation requires the exponent to be a multiple of 3:

$$\boxed{1 \leq |a| < 1000, \quad n = 3k}.$$

Thus,

$$47\,000\,000 = \underbrace{4.7 \times 10^7}_{\text{scientific}} = \underbrace{47 \times 10^6}_{\text{engineering}} = \underbrace{47 \text{ M}}_{\text{with an SI prefix}}.$$

Scientific notation is particularly useful for mathematical calculations and comparisons of

TO THE READER:

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